Path Independence SOLNS

Problem 1

a) $\vec{F} = 2x\vec{i} + 3y\vec{j} + 4z\vec{k}$

conservative

$\Rightarrow \text{curl}\,\vec{F} = \vec{0}$

b) $\vec{F} = [y, x+z, -y]$

$\Rightarrow \text{curl}\,\vec{F} = [-2, 0, 0] \neq \vec{0}$

c) $\vec{F} = -y\vec{i} + x\vec{j}$

$\Rightarrow \text{curl}\,\vec{F} = [0, 0, 2] \neq \vec{0}$

d) $\vec{F} = [e^{y+2z}, xe^{y+2z}, 2xe^{y+2z}]$

conservative

$\Rightarrow \text{curl}\,\vec{F} = [0, 0, 0] = \vec{0}$

Problem 2

a) $[2x, 3y, 4z] = [f_x, f_y, f_z]$

$f_x = 2x \Rightarrow f = x^2 + g(y,z)$

$3y = f_y = g_y \Rightarrow g = \frac{3}{2}y^2 + h(z)$

$f = x^2 + \frac{3}{2}y^2 + h(z)$

$4z = f_z = h_z \Rightarrow h = 2z^2 + c$

$$\boxed{f = x^2 + \frac{3}{2}y^2 + 2z^2}$$

b) $[e^{y+2z}, xe^{y+2z}, 2xe^{y+2z}] = [f_x, f_y, f_z]$

$f_x = e^{y+2z} \Rightarrow f = xe^{y+2z} + g(y,z)$

$xe^{y+2z} = f_y = xe^{y+2z} + g_y \Rightarrow g_y = 0 \Rightarrow g = h(z)$

$f = xe^{y+2z} + h(z)$

$2xe^{y+2z} = f_z = 2xe^{y+2z} + h_z \Rightarrow h_z = 0 \Rightarrow h = C$

$$\boxed{f = xe^{y+2z}}$$